



AI-POWERED EDUCATIONAL CHATBOT



INTRODUCTION



1. **AI-Powered Educational Chabot** designed to provide accurate answers of student's educational questions related to Data Science.
2. It understands the meaning of the user's question using Sentence Transformers and semantic similarity from a knowledge database containing question-answer pairs.
3. Context memory helps the chatbot understand follow-up questions and maintain conversation continuity otherwise fallback answer.
4. Provides an interactive Stream lit Chabot with persistent SQLite chat history.



OBJECTIVE

1. To develop an intelligent Chabot for **educational question answering**.
2. To retrieve relevant answers using **Sentence Transformer embedding , Cosine Similarity and Context-memory**.
3. To provide a **user-friendly Stream lit interface** with suggested questions and similarity scores.
4. To maintain **persistent chat history using SQLite** across refreshes and application restarts.





METHODOLOGY

1. **Dataset Collection:** Collect educational question-answer pairs in a structured CSV knowledge database.
2. **Data Cleaning:** Clean and preprocess the knowledge base by removing duplicates and invalid data.
3. **Model Embedding (Sentence Transformer):** Use all-MiniLM-L6-v2 to understand the semantic meaning of questions and convert all knowledge-base questions or user's questions into numerical vector embedding.
4. **Cosine Similarity:** Calculate cosine similarity between the user query and stored question embedding's.
5. **Context Memory:** Use context memory to understand follow-up questions and maintain the active topic.
6. **Threshold & Fallback:** Return the answer if the score meets the threshold; otherwise, show a fallback response.
7. **Chat & Storage:** Display the response through Stream lit and save conversation history using SQLite.





TOOLS & TECHNOLOGY

1. **Python** – Main programming language used to develop the Chabot.
2. **Pandas** – Used for CSV loading, data cleaning, and dataset handling.
3. **NumPy** – Used for numerical operations and finding the highest cosine similarity score.
4. **Sentence Transformers(all-MiniLM-L6-v2)** – Used for generating semantic embedding.
5. **Memory** – Context Memory used for follow-up question retrieval or Session State used for storing chat timestamp.
6. **Streamlit** – Used to create the interactive Chabot web application.
7. **SQLite** – Used for persistent storage of chat history.
8. **CSV** – Used as the educational knowledge-base format.
9. **VS Code / Python Environment** – Used for project development and testing.





FUTURE ENHANCEMENT

- 1. Larger Knowledge Base & Document Retrieval**
– Add PDFs, notes, textbooks, and other educational documents.
- 2. Hybrid Search & RAG** – Combine semantic search with keyword search and Retrieval-Augmented Generation for better answers.
- 3. Multilingual & Voice Support** – Allow students to ask questions in multiple languages using voice input.
- 4. Cloud Deployment & User Management** – Add authentication, cloud database, analytics, and deploy the Chabot for multiple users.





THANK YOU